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http://www.ellsbury.com/enigmabombe.htm

The Turing Bombe

What it was and how it worked

The King hath note of all that they intend,
By interception which they dream not of.

William Shakespeare. *Henry V.*

1. Cribs and Menus

It will be understood from the nature of the Enigma machine that, having been given a replica Enigma by the Poles, complete with a set of rotors and their internal wiring, the task confronting Dillwyn Knox and Alan Turing, the primary British cryptanalysts, was to determine the particular settings of the Enigma machine used to encipher a particular message. Turing and Knox considered three possible methods of attack:

1. Ciphertext-only analysis. This method was later employed in Heath Robinson and the Robinson family of machines and the Colossus series of primitive computers against the FISH traffic encrypted by the Geheimschreiber. It requires ciphertext of a substantial length and so was considered impractical for Enigma traffic which typically consisted of messages of less than 200 characters which is far too little for this type of attack.

2. Discriminant attack. This method, based on an extraordinary German procedural error, was exploited to great effect by the Poles, who achieved the remarkable feat of reconstructing the internal wiring of the rotors by studying the repeated encryption of the three-letter discriminant. It was considered too fragile by Turing and Knox in that the Germans might at any moment change their operational procedures and thereby render useless overnight any machines constructed to assist in the attack. This decision was wise for the Germans did indeed change their procedural methods on May 1st 1940.

3. Probable-phrase attack. This method was based on Turing's realisation that it is possible to exploit the relationship between a known, or guessed, portion of plaintext, called a crib, and the ciphertext. In particular, Turing realised that the

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relationship between the crib and the ciphertext would preclude some of the stupendous number of ways in which the Enigma machine could have been set up to encipher the plaintext and that accordingly it might be possible to construct a machine to eliminate from consideration many of the ways in which the Enigma could have been set up.

Even if such a machine proved to be practical, would it be possible to find cribs in sufficient quantity? It was not necessary to find a crib for each message, rather, what was required was one crib for each network each day. For once the Enigma settings had been found for that net for that day every message on that net on that day could be deciphered. The Germans would then typically change the settings at midnight and a new crib would have to be found the following day.

But how might cribs be found? German military communications were often expressed in a stereotypical manner so that it was not necessarily difficult to guess a particular phrase that might appear in the message. Further, it might be possible to determine its exact position in the ciphertext by exploiting the property of the Enigma machine which ensured that it never encoded a letter as itself. One very fruitful source was weather reports, for example, the crib used for the first break on D-Day was:

WETTERVORHERSAGEBISKAYA

Now suppose that the ciphertext includes the sequence:

…QFZWRWIVTYRESXBFOGKUHQBAISEZ…

To determine which letter of the crib corresponds to which letter of the ciphertext we slide the crib alongside the ciphertext until we find a correspondence where the same letter does not appear in the same position in the crib and the ciphertext.

Q F Z W R W I V T Y R E **S** X B F O G K U H Q B A I S E Z
W E T T E R V O R H E R **S** A G E B I S K A Y A

Figure 1.1

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For example, in Figure 1.1, the first S in WETTERVORHER**S**AGEBISKAYA corresponds to the first S in …QFZWRWIVTYRE**S**XBFOGKUHQBAISEZ…, so we know that this cannot be the correct correspondence because the Enigma machine has the property that it can never encipher a letter as itself.

Q F Z W R W I **V** T Y R **E** S X B F O G K U H Q B **A** I S E Z
W E T T E R **V** O R H **E** R S A G E B I S K A Y **A**

Figure 1.2

So we slide the crib along the ciphertext by one character and again check the correspondences. In this example we again see in Figure 1.2 that the crib is still in the wrong position because a V, E and A appear in both the crib and the ciphertext in the same position.

Q F Z W R W I V T Y **R** E S X B F O G K U H Q B A I S E Z
W E T T E R V O **R** H E R S A G E B I S K A Y A

Figure 1.3

So again we slide the crib along, and again we fail to find a possible correspondence because in Figure 1.3 R appears in the crib and the ciphertext in the same position and because in Figure 1.4 W, G and A appear in both the crib and the ciphertext in the same position.

Q F Z **W** R W I V T Y R E S X B F O **G** K U H Q B **A** I S E Z
W E T T E R V O R H E R S A **G** E B I S K **A** Y A

Figure 1.4

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Q F Z W R W I V T Y R E S X B F O G K U H Q B A I S E Z
W E T T E R V O R H E R S A G E B I S K A Y A

Figure 1.5

In Figure 1.5 we at last see how the crib can be matched to a sequence of cipher text, so there is the possibility that

RWIVTYRESXBFOGKUHQBAISE

is the encryption of

WETTERVORHERSAGEBISKAYA

for one of the 60 rotor orders, 17576 rotor positions, 676 Ringstellung positions and 150,000,000,000,000 stecker swappings.

The question now is which one of the 107,458,687,327,300,000,000,000 possible rotor orders, rotor positions, Ringstellung and stecker swappings was used to encipher the message?

The crib and ciphertext can be paired and numbered as follows:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
R	W	I	V	T	Y	R	E	S	X	B	F	O	G	K	U	H	Q	B	A	I	S	E
W	E	T	T	E	R	V	O	R	H	E	R	S	A	G	E	B	I	S	K	A	Y	A

So, if the crib is correctly matched with the ciphertext, we know that some setting of the Enigma will encode R as W and W as R in a particular position which we will call position 1, the Enigma rotor(s) step and then in the next position, which we call position 2, it will encode W as E and E as W, then the rotor(s) step and then in the next position which we call position 3, it will encode I as T and T as I, and so on.

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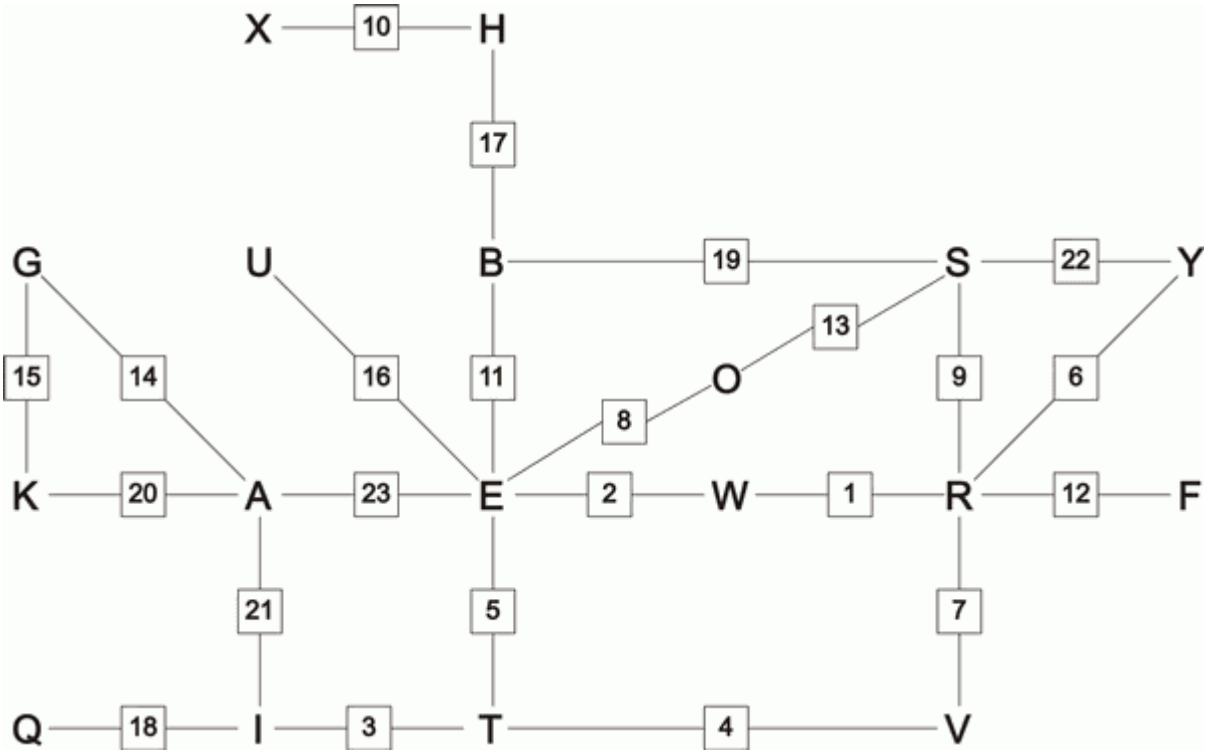


Figure 1.6 Crib-Ciphertext Pairings

These pairing relationships can be diagrammed as in Figure 1.6. In this diagram the square boxes represent scramblers and the numbers in the boxes represent the positions of the scramblers relative to the position of the scrambler which transformed the first pairing.

Recall that each transformation of one letter into another by the Enigma consists of:

a stecker swapping, followed by
a scrambler encryption, followed by
another stecker swapping.

Turing realised that some of these transformations have implications for other transformations and thus can be fed back into themselves to confirm consistency, or, more decisively, to demonstrate a contradiction.

For example, let us test the hypothesis that E is steckered to K. Now consider how E is transformed into A at relative position 23.

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If E is steckered to K, K will be input to the scrambler at relative position 23 which will output some other letter v_1. Since we know that E transforms into A at relative position 23, we know that A must be steckered to v_1.

But if A is steckered to v_1, v_1 will be input to the scrambler at relative position 21 which will output some other letter v_2. Since we know that A transforms into I at relative position 21, we know that I must be steckered to v_2.

But if I is steckered to v_2, v_2 will be input to the scrambler at relative position 3 which will output some other letter v_3. Since we know that I transforms into T at relative position 3, we know that T must be steckered to v_3.

But if T is steckered to v_3, v_3 will be input to the scrambler at relative position 5.

Now consider the output of the scrambler at relative position 5 when it receives v_3 as its input. Since we know that T transforms into E at relative position 5, we know that the output letter of the scrambler at relative position 5 must be steckered to E.

Suppose, when given v_3 as its input, the scrambler at relative position 5 outputs J. By our original hypothesis E is steckered to K, but by our chain of implications E is steckered to J, so in this case E is steckered to both K and J. But the construction of the Enigma does not permit one letter to be steckered to two letters, so E cannot be steckered to both K and J, so our original hypothesis has resulted in an impossible consequence and so must be false.

But now suppose, when given v_3 as its input, the scrambler at relative position 5 outputs K. Then by our original hypothesis E is steckered to K, and by our chain of implications E is steckered to K, so in this case E is steckered only to K, thus our original hypothesis has resulted in a consistent consequence and so may be true.

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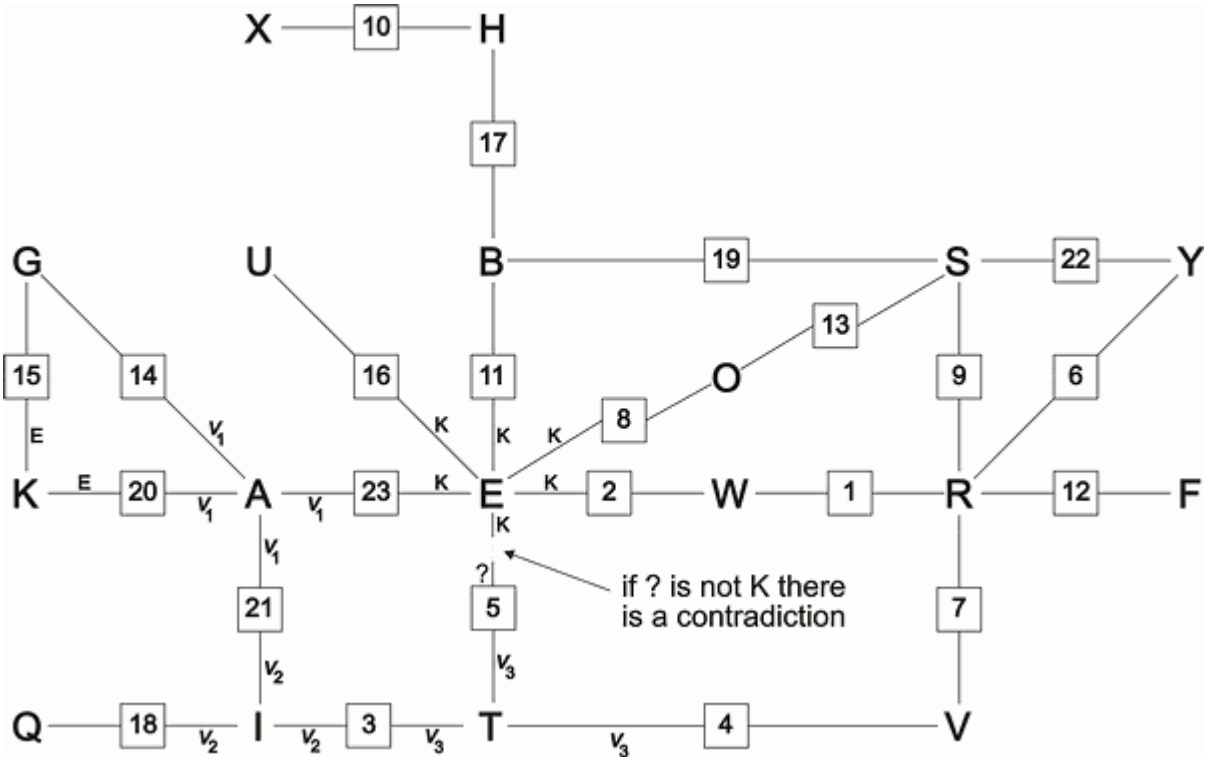


Figure 1.7 Testing the Hypothesis that E is Steckered to K

Figure 1.7 shows the hypothesis that E is steckered to K looping back via the chain of implications we have considered.

In general, a steckering hypothesis is consistent if and only if, having returned via some chain of implications to its origin, it implies itself.

Note that by using this method we need loops to feedback into our original hypothesis. When there are four loops in the crib-ciphertext pairing, the feedback imposes a very severe consistency condition, which the vast majority of rotor orders and rotor positions are unable to satisfy.

This concept, of using loops in the crib-ciphertext pairing to test for inconsistencies is the fundamental concept of Turing's probable-phrase attack on the Enigma.

Turing realised that it was possible to represent loops of implications with electrical circuitry and that therefore it was possible to mechanise the search for those rotor orders and positions which satisfy the consistency conditions. The machine which he designed to perform this task is called the Turing bombe.

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In other words, the bombe checked whether, with the current rotor order, the current rotor position and any stecker swapping, the crib and ciphertext could be transformed into each other. Thus by successively checking the 60 rotor orders each of 17,576 positions, the bombe performed an exhaustive search of the

107,458,687,327,300,000,000,000

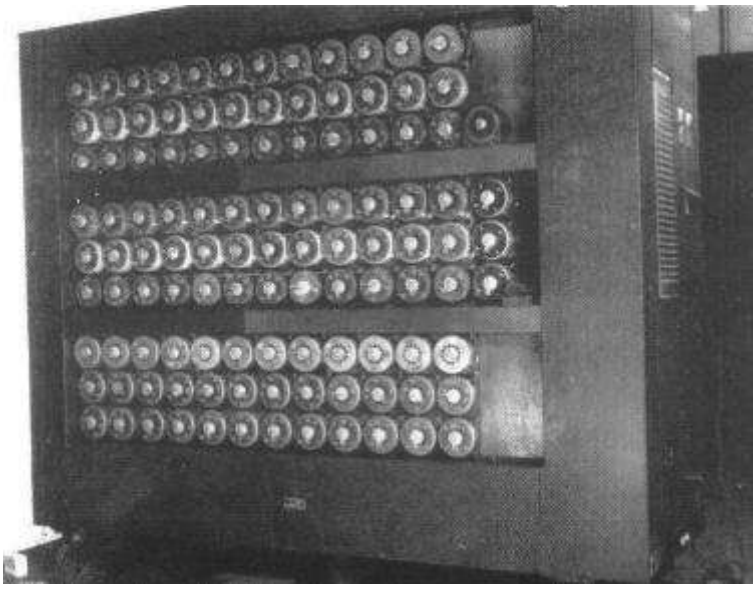
ways in which the Enigma machine could have been set up. By this means it located the needle in the haystack, or perhaps more appropriately, the key to the Third Reich.

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The Turing Bombe

What it was and how it worked



The Turing Bombe

2. Description of the Bombe

Although Turing is the father of the British bombe, the engineering, as distinct from logical, design of the bombe was due to Harold 'Doc' Keen an employee of The British Tabulating Machine Company at Letchworth which built the bombes under the Cantab trade-name. The bombe weighed about one ton, was housed in a bronze-coloured metal cabinet about 7 feet wide, 6 feet 6 inches tall and 2 feet deep and was mounted on castors. Protruding from the front of the cabinet there were 108 shafts (more in some models, fewer in the two prototypes) arranged in three 12 x 3 arrays on which drums were mounted.

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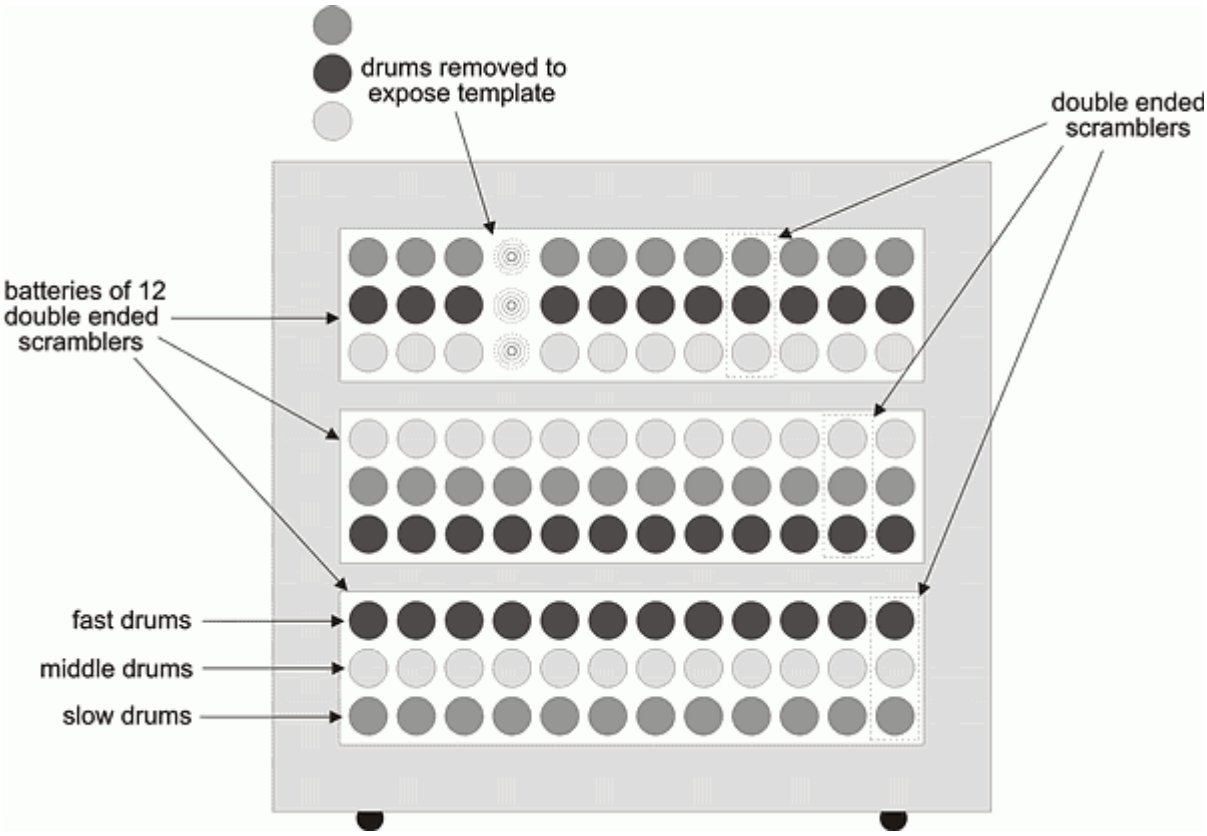
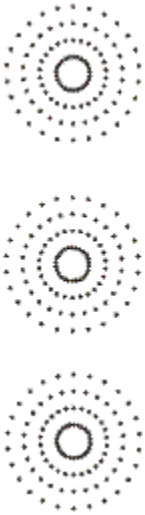


Figure 2.1 The Front Panel of a Bombe

See the above photograph of a British bombe and Figure 2.1, a diagram of the front panel of an idealised bombe.

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On the front panel of the bombe, around each shaft, there were 104 electrical contacts arranged in four concentric circles of 26 contacts each. The contacts associated with each vertical triplet of shafts formed a template of 312 contacts as indicated in the diagram to the left.

These contacts were connected by wiring behind the template as follows:

each contact in the	**was connected to the corresponding contact in**
outer (fourth) ring of the top set	a 26-way cable which could be plugged-up elsewhere in the bombe
third ring of the top set	the outer ring of the middle set
third ring of the middle set	the outer ring of the bottom set
third ring of the bottom set	the second ring of the bottom set via wires which duplicated the Umkehrwalze
inner ring of the bottom set	the second ring of the middle set
inner ring of the middle set	the second ring of the top set
inner ring of the top set	a 26-way cable which could be plugged-up elsewhere in the bombe

Each vertical triplet of drums, the template they were mounted on, and the connecting wiring behind the template, constituted a double-ended scrambler. The task of each double-ended scrambler was to perform exactly the same

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transformation as that performed by the scrambler in the Enigma machine but unlike the single-ended Enigma scramblers, the bombe's scramblers had to be double-ended. For reasons that will appear later, it was necessary to connect a number of scramblers in series, that is, it was necessary to connect scrambler a to scrambler b, and scrambler b to scrambler g, without directly connecting scrambler a to scrambler g, but single-ended scramblers cannot be connected in this way.

Although each drum performed exactly the same transformation as its corresponding Enigma rotor, to enable the bombe's scramblers to be double-ended each drum had two sets of internal wiring each of which was configured to match the internal wiring of one of the rotors of the Enigma machine. For example, the drum corresponding to rotor IV of the Enigma would have 2 sets of internal wires, each set corresponding to the internal wiring in rotor IV. The task of the wiring in the drums was to bridge the contacts of the outer and third rings, and of the inner and second rings of contacts. Each double-ended scrambler was symmetrical, for example, if R was input at the left end of the scrambler resulting in F being output at the right end, then if R was input at the right end, F would be output at the left end.

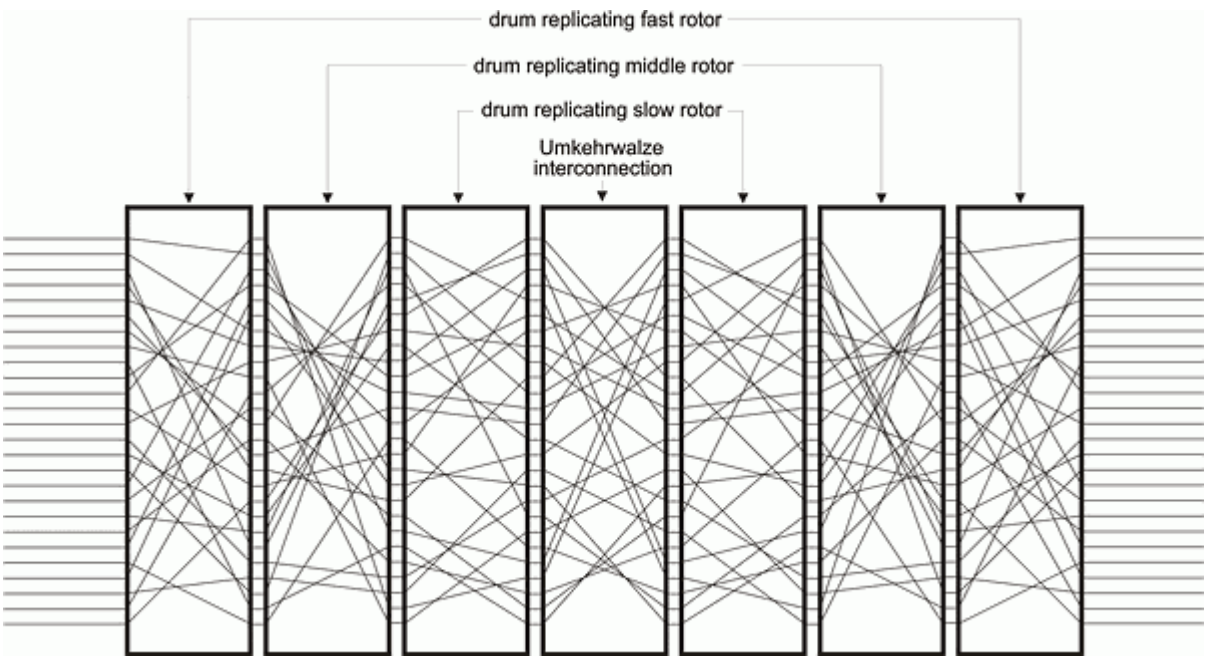


Figure 2.2 Electrical Pathways Through a Double Ended Scrambler

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Figure 2.2 shows the logical pathways through a double-ended scrambler.

The drums were approximately 5" in diameter and had 104 electrical brushes arranged in four concentric circles of 26 brushes each to match the contacts on the template. The operators had to take great care to ensure that the brushes did not short against each other. Each drum was coloured in accordance with the rotor that it replicated, so that, the drums would be coloured, say, red for rotor I, blue for rotor II, black for rotor III, and so on. All 26 letters of the alphabet were printed around the circumference of each drum so that an operator could readily determine its orientation.

Each of the three rows of double-ended scramblers formed a battery of scramblers. Each of the three batteries in the bombe was electrically isolated from the other two. So a bombe consists of three electrically isolated but mechanically connected batteries. Inside the bombe there was an electric motor which via a belt, a number of shafts, and an arrangement of gears, rotated the shafts, and their attached drums, in precise synchrony.

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The Turing Bombe

What it was and how it worked



Alan Turing
Designed the bombe



Gordon Welchman
Discovered the diagonal board

3. How the Bombe was Plugged Up

Due to its complexity, the circuitry of the bombe does not readily lend itself to being diagrammed. But to obtain a clear understanding of how the bombe was plugged up and how it worked, it is advantageous to examine a circuit diagram for the entire bombe. For the real bombe, this is impractical without resorting to large-scale diagrams, but there is an alternative, we can conveniently diagram the complete circuitry of a simplified bombe.

We shall accordingly consider a simplified bombe designed to search for keys for a simplified Enigma with an alphabet of only eight letters: A, B, C, D, E, F, G and H. The principles by which the simplified bombe operates are identical to those of the real bombe.

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Suppose we have obtained the crib BEACHHEAD which we have paired, as described earlier, with the ciphertext EDBGEAHDB as follows:

1	2	3	4	5	6	7	8	9
B	E	A	C	H	H	E	A	D
E	D	B	G	E	A	H	D	B

Then, also as described earlier, we can derive the crib diagram:

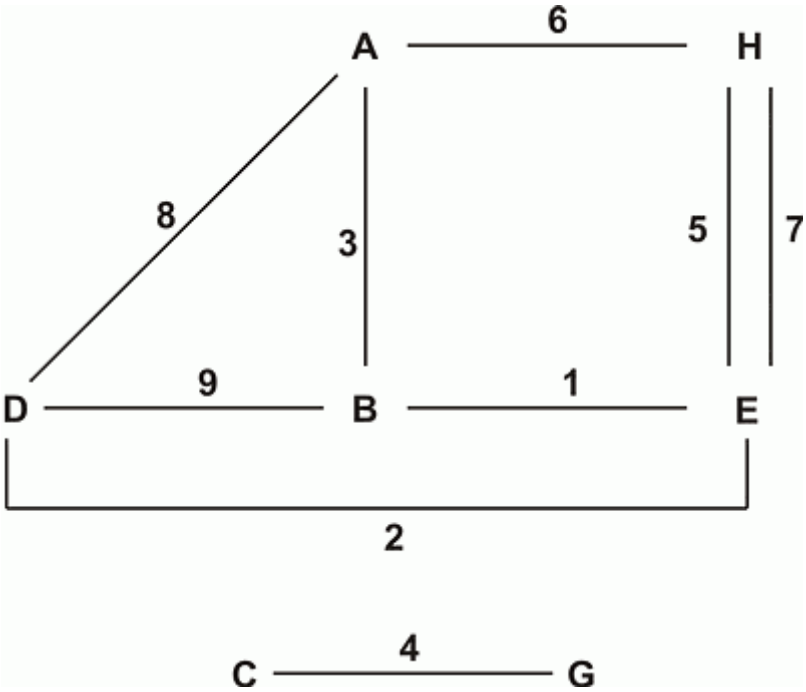


Figure 3.1 Menu for the Simplified Enigma

This diagram, which is called a menu, tells us all we need to know to plug-up the bombe.

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After Turing had designed the bombe, Gordon Welchman, a colleague of Turing's at Bletchley Park, realised the reciprocal property of the stecker meant that if, for example, W was steckered to Q then Q must be steckered to W, and that it was possible to reflect this property in a device which came to be called the diagonal board.

The diagonal board consisted of 26 cables each containing 26 wires leading into and out of it, with a total of 325 connections between them. For example, the *a* wire in the *B* cable was permanently connected to the *b* wire in the *A* cable, and the *a* wire in the *C* cable was permanently connected to the *c* wire in the *A* cable, and so on.

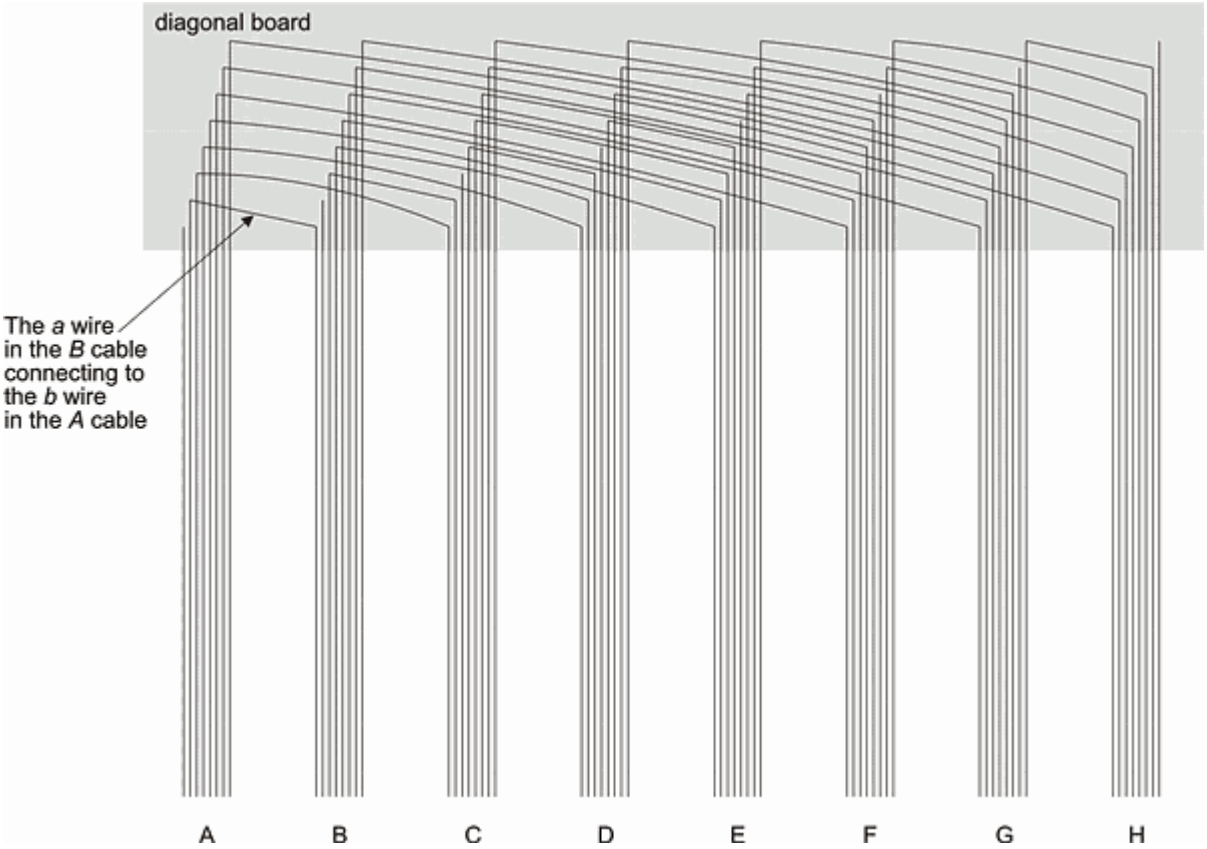


Figure 3.2 Diagonal Board of the Simplified Bombe

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As shown in Figure 3.2, the diagonal board of our simplified bombe has 8 cables each containing 8 wires.

The effect of the diagonal board, when attached to the bombe, was to increase the feedback into the double-ended scramblers and thus reduce the necessity of obtaining loops in the crib-ciphertext pairing. This in turn made it possible to use shorter menus which were less likely to include a turnover of the Enigma's middle rotor during the encipherment of the plaintext. This is highly advantageous for reasons that will appear later.

The two ends of a double-ended scrambler were plugged into the two cables of the diagonal board which represented the two letters connected by the scrambler in the menu. In other words, each scrambler bridges two of the diagonal board's cables. When all the scramblers had been connected in this way the drums were rotated into positions in accordance with their relative positions shown in the menu.

In our example menu, B is connected to E at relative position 1, so one end of a double-ended scrambler is plugged into the diagonal board's cable which represents B, and the other end of the same double-ended scrambler is plugged into the diagonal board's cable which represents E. Similarly, a second double-ended scrambler is plugged to bridge D and E; a third to bridge A and B; a fourth, C and G; a fifth, E and H; a sixth, A and H; a seventh, E and H again; an eighth, A and D; then the ninth and final double-ended scrambler, bridges B and D.

Next we mount the drums on their shafts. We first select a particular rotor order to try. Suppose we try rotors I, II and III. We mount nine drums each of which corresponds to rotor I on the first nine top shafts in a battery, then we mount nine drums each of which corresponds to rotor II on the first nine middle shafts in the same battery, then we mount nine drums each of which corresponds to rotor III on the first nine lower shafts in the same battery.

The drums must now be rotated to their appropriate positions. So we rotate each drum until it has the letter A painted on its circumference in the 12 o'clock position. To reflect their relative positions in the crib-ciphertext pairing the drums are now reset in accordance with the following table:

drum	cables connected by drum's scrambler	scrambler position in crib-ciphertext pairing	drums letter in 12 o'clock position
top	B and E	1	A
top	D and E	2	B

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top	A and B	3	C
top	C and G	4	D
top	E and H	5	E
top	A and H	6	F
top	E and H	7	G
top	A and D	8	H
top	B and D	9	A
middle	B and D	9	B

This completes the plugging-up of a single battery.

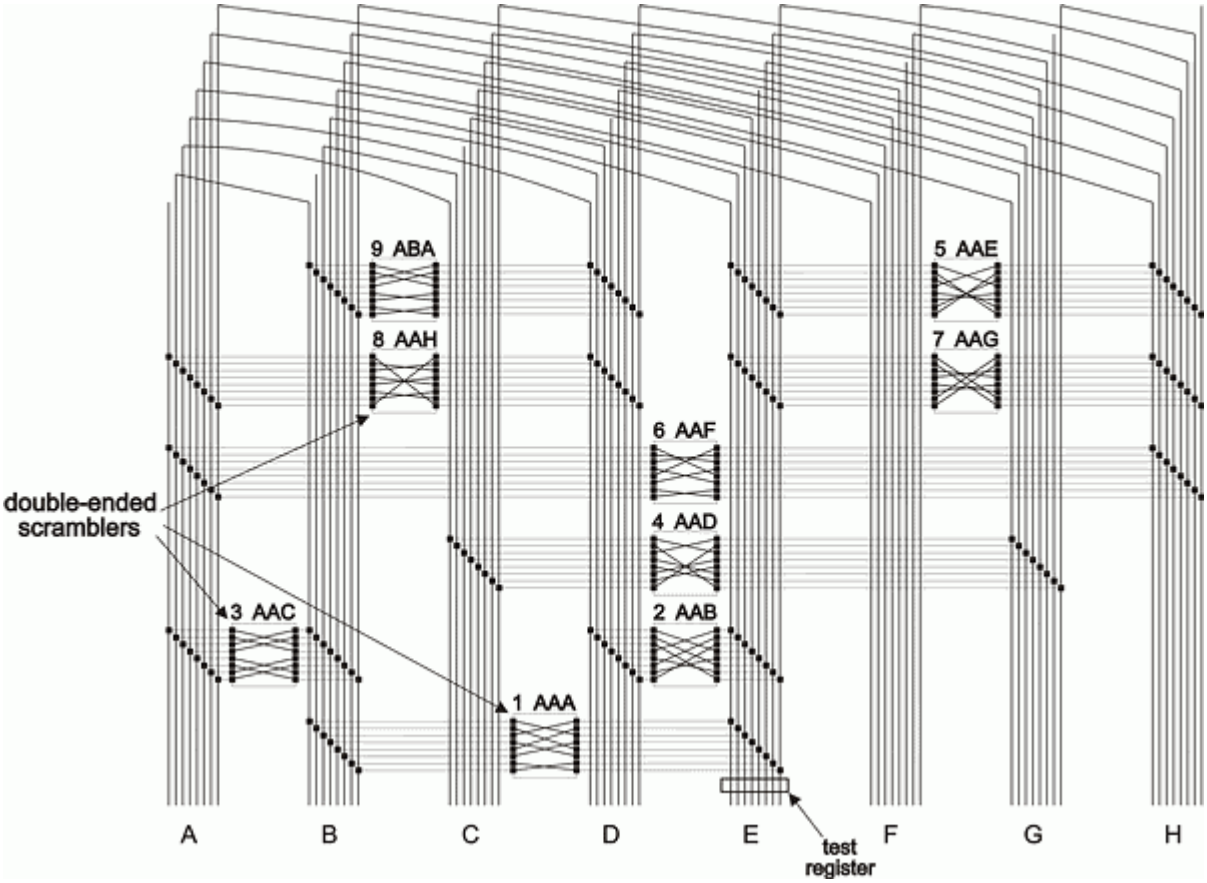


Figure 3.3 Plugged-up Simplified Bombe

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See Figure 3.3

The other two batteries are now plugged up in exactly the same way except that different drum orders are selected.

Note that the loops in the menu are represented by loops in the plugged-up bombe.

The bombe is now fully plugged up and ready to run.

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The Turing Bombe

What it was and how it worked

It is an old maxim of mine that when you have excluded the impossible,
whatever remains, however improbable, must be the truth.

Sir Arthur Conan Doyle. *The Beryl Coronet.*

4. How the Bombe Worked

When the bombe was set in motion the drums in the upper row of each of the three batteries rotated in precise synchrony at 120 rpm. When they had completed one revolution the middle row of drums in each battery rotated to the next position. That is, they rotated 1/26 of one revolution, and when they had rotated through one revolution the lower row of drums rotated to the next position. This continued until either the bombe detected a 'stop' or until the drums had returned to their original positions.

A stop is a setting where the transformations determined by the rotor order, rotor position, and some stecker swapping are consistent with the transformations that yielded the ciphertext from the crib. But how did the bombe detect when the consistency conditions had been met? In other words, how did it know when to stop?

A test register was connected to one of the cables, typically the cable representing the letter that appeared most frequently in the menu, so that each of the 26 wires in the cable led into and out of the test register. Each wire was in one of two states which we shall call live and dead, and the register was capable of distinguishing how many of its wires were live.

When the bombe was started it applied a voltage to an arbitrary wire in the test register's cable, say the *a* wire in the *E* cable, to represent the hypothesis that A

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is steckered to E and E is steckered to A. Every wire connected to the a wire in the *E* cable and the *e* wire in the *A* cable would immediately become live, but suppose that another wire in the *E* cable, for example the *h* wire, becomes live. This represents the hypothesis that H is steckered to E and E is steckered to H. But E cannot be steckered to both A and H so we have arrived at a contradiction and we know that this rotor order, with this rotor position and the hypothesis that A is steckered to E is false.

Do we now need to apply a voltage to the *b* wire in the *E* cable to test the hypothesis that B is steckered to E and E is steckered to B? We do not.

Having applied the current to the *a* wire in the *E* cable and therefore the *e* wire in the *A* cable and having had the current return to the *h* wire in the *E* cable and therefore the *e* wire in the *H* cable, we note that the current will pass through any scrambler connected to the *A*, *E* and *H* cables to wires in other cables which will not generally be the wires already energised in the *A*, *E* and *H* cables. Thus as the current flashes through the cryptanalytical circuitry of the bombe feeding back into cables and scramblers, more and more wires become live until, with a good menu, the feedback will almost always make all the wires in the test register live. There are two notable exceptions.

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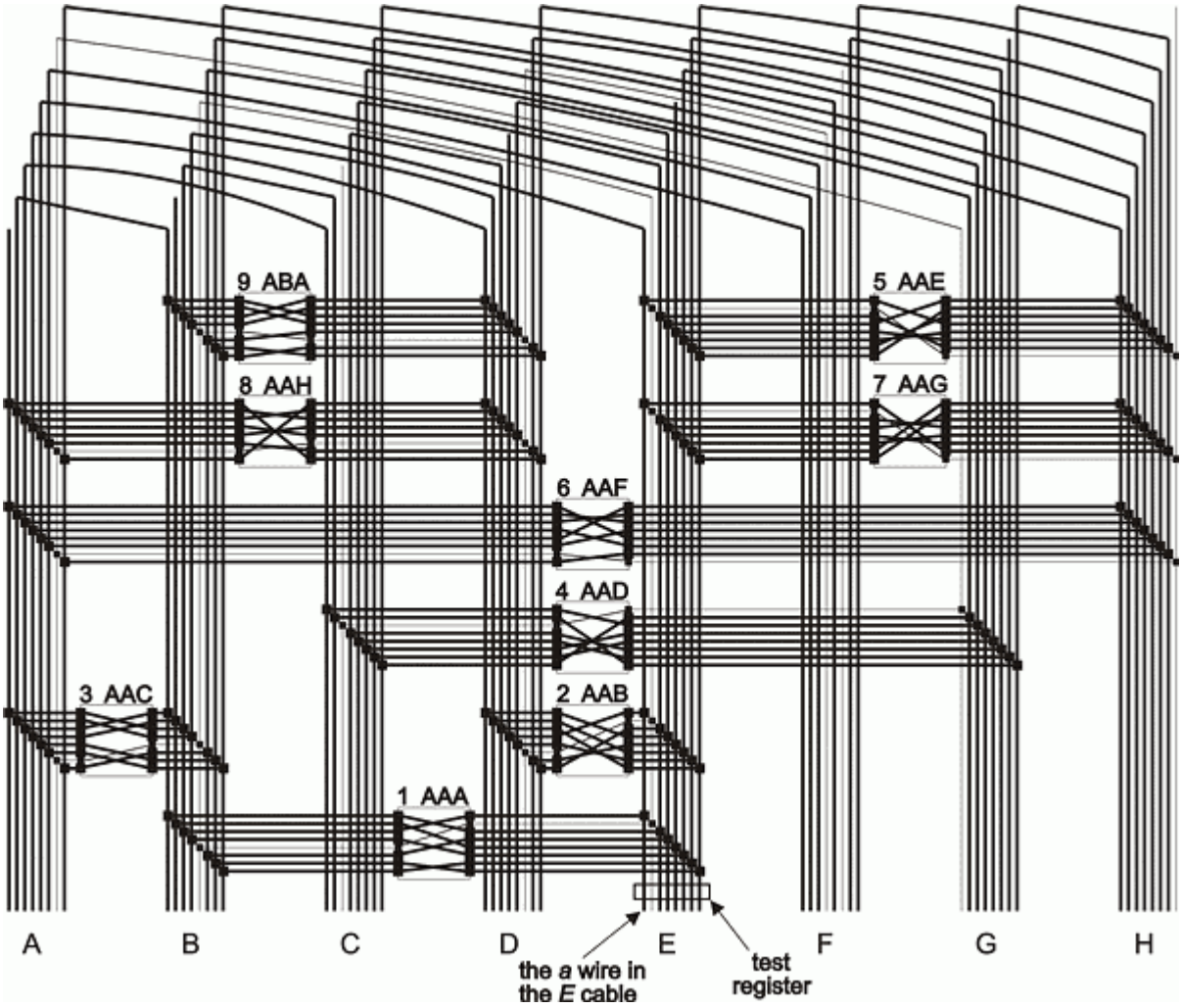


Figure 4.1 Simplified Bombe with 25 Live Wires in the Test Register

Suppose that the rotors are in the correct position but that the hypothesis being tested is false.

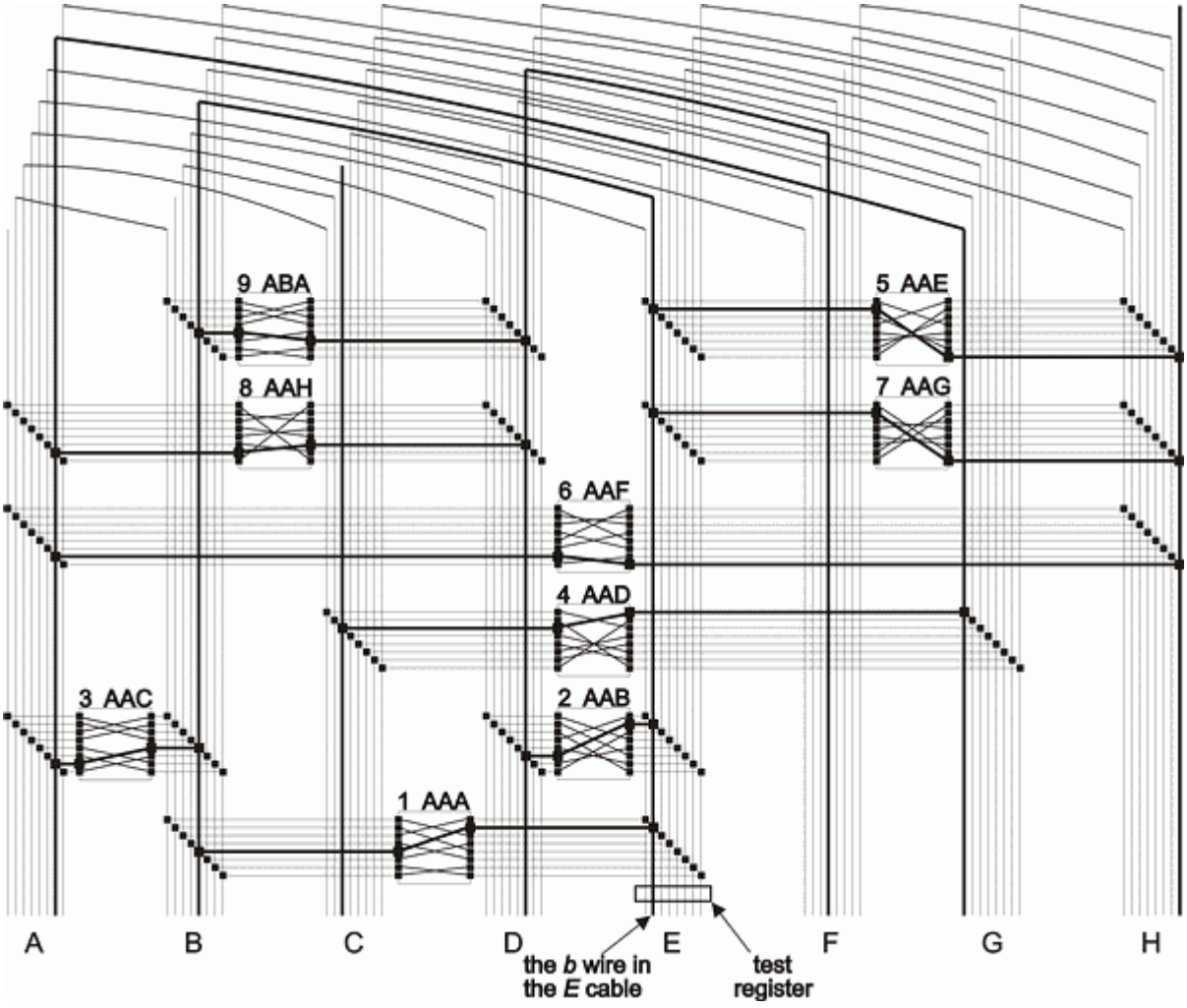
For example, suppose that the rotor order and position is correct and that the false hypothesis that E is steckered to A is being tested. This situation is shown in Figure 4.1 in which the thick black lines indicate those wires which are live after a voltage has been applied to the *a* wire in the *E* cable.

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In this case the wires representing the true hypotheses are not connected to the wires representing the false hypotheses. But, for the reason discussed above, the wires representing the false hypotheses are connected together so when any one of them has a voltage applied to it, the voltage appears on all of them.

So, in this case, if we look in the test register we will find exactly 25 live wires and one dead wire. The dead wire, and all the wires connected to it, indicate stecker swapping consistent with the crib-ciphertext pairing.

In the example shown in Figure 4.1, the dead wires in the cables indicate that A is steckered to G, B is steckered to E, C is self-steckered, D is steckered to F, and H is self-steckered.



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Figure 4.2 Simplified Bombe with 1 Live Wire in the Test Register

Now suppose that the rotors are in the correct position and that the hypothesis being tested is true.

For example, suppose that the rotor order and position is correct and that the true hypothesis that E is steckered to B is being tested. This situation is shown in Figure 4.2, which has exactly the same scrambler positions as Figure 4.1 and in which the thick black lines again indicate those wires which are live, this time after a voltage has been applied to the *b* wire in the *E* cable.

In this case the wires representing the true hypotheses are not connected to the wires representing the false hypotheses so when a voltage is applied to any one of the wires representing a true hypothesis, the voltage appears on all the other wires which also represent true hypotheses, and on no other wires.

So, in this case, if we look in the test register we will find exactly 1 live wire and 25 dead wires. The live wire, and all the wires connected to it indicates stecker swapping consistent with the crib-ciphertext pairing.

In the example shown in Figure 4.2, the live wires in the cables indicate that A is steckered to G, B is steckered to E, C is self-steckered, D is steckered to F, and H is self-steckered, in other words, the steckering that the bombe has found is identical to that of Figure 4.1. The wires which are live in the test register in these two situations are the logical complement of each other.

So there are two stop conditions which the bombe looks for, either exactly one live wire or exactly 25 live wires in the test register.

In *The Hut Six Story* Gordon Welchman confirms that the stop condition for the bombe is that given above. However it will sometimes, but very rarely, occur that there are more than 1 but less than 25 live wires in the test register. We cannot discount the possibility that this represents a genuine stop so I think it was likely that the bombe stopped whenever there were less than 26 live wires in the test register. If the bombe stopped when there were, say, 24 live wires in the test register, a Wren could have performed a check by putting a voltage on one of the dead wires and then noting whether the other dead wire became live. If so, then we know the wires are connected and therefore cannot represent a solution, but if not then both wires represent potential solutions. Occurrences of this kind were exceedingly rare.

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Note that in the example in Figure 4.1 and Figure 4.2, the *E* and *H* cables are bridged by two scramblers. Since these scramblers are in two different positions, a current input on, say, the *g* wire in one of them will not generally yield the same output as a current input on the *g* wire in the other. Sometimes however, the current input on the *g* wire in one of them will yield the same output as a current input on the *g* wire in the other, even though the two scramblers will yield different outputs from other identical inputs. This will occur, on average, in one position in 26. This situation is illustrated in Figures 4.1 and 4.2. In the other 25 out of 26 positions, the current passes through one scrambler to energise a wire in one of the two cables connected to the other scrambler and is fed back by the other scrambler through the other cable, to the former scrambler. This feedback will usually continue until all 26 wires in both the cables are live. Therefore two scramblers, when bridging the same two cables, are a very potent means of disposing of steckering hypotheses.

When the bombe detected a stop it recorded the position of the upper row of drums on one of 26 relays. The turning of the middle and lower rows of drums was inhibited and the motor allowed to slow down. The upper row of drums was then rotated into the position recorded by the relays. The bombe thus halted in the correct position and the stop position was noted by the operator and the bombe was restarted.

The stop was checked to determine whether it was correct or merely a rotor order and position that happened to be consistent with the crib and ciphertext but which was not the correct key for the Enigma.

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The Turing Bombe

What it was and how it worked

5. Checking the Stops

Some checks were performed by a device called a machine gun which searched the diagonal board for two letters steckered to the same third letter. For example, if, say, G is steckered to Y, and R is steckered to Y then due to the reciprocal nature of the Enigma's plugging, Y must be steckered to both R and G, but this is impossible so any stop exhibiting this property can be rejected. The machine gun got its name from its array of uniselectors which sounded like a burst of machine gun fire as they successively searched the cables of the diagonal board for duplicate steckers. The bombe had to be halted to use the machine gun and as it would not eliminate all false stops it was not always used.

Note that in Figure 4.1, the presence of two dead wires in the *F* cable does not imply that F is doubly steckered. This is because F is not in the menu and therefore the *F* cable has no scramblers attached to it so the only way its wires can be energised is through the diagonal board. Since the *f* wire in the *F* cable is not connected to any other cable via the diagonal board it is not connected to any other cable at all and therefore can never be energised.

The check was usually performed by a Wren on a checking machine, which was similar to an Enigma with the stepping mechanism disabled. The rotor order and positions were set up in accordance with the order and positions discovered by the bombe. The stecker pairings discovered by the bombe were also plugged up on the checking machine, these would not generally be all the necessary stecker pairings however so any additional steckers had to be determined. There also remained the tedious task of manually determining the Ringstellung.

When the checking machine was set up in accordance with the stop the ciphertext was keyed in, and if it yielded German text, or a close approximation to it, that indicated the stop was correct and the cry *'Job's up, strip machine!'* was called out and the bombe which found the stop, and all the other

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bombes working to break the same key were halted and then unplugged in preparation for the next job.

If this procedure failed to yield German text the stop was not correct and the next stop was examined.

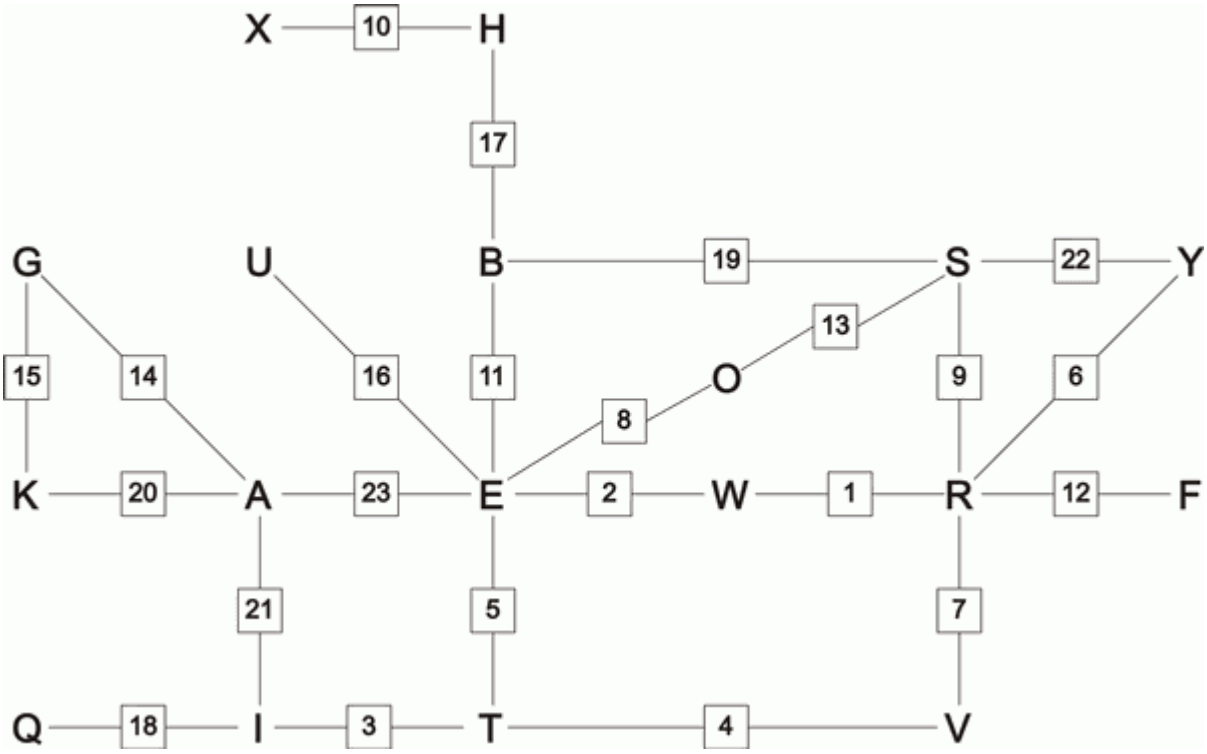


Figure 1.6 Crib-Ciphertext Pairings

Returning now to Figure 1.6 we see that it encompasses a span from scrambler position 1 to position 23 of 23 letters. There is therefore a 23/26 or 88% chance of the Enigma's middle rotor turning over during encipherment. If this occurred some of the bombe's scramblers would be in the wrong position relative to the other scramblers and the bombe will therefore not produce valid stops. So to eliminate the possibility of a middle rotor turnover ruining the bombe run we can form two separate menus, one using the span 1-13, the other using the span 14-23. We know that at most one of these two menus can have suffered a middle rotor turnover during its encipherment, so at least one of them will yield valid stops.

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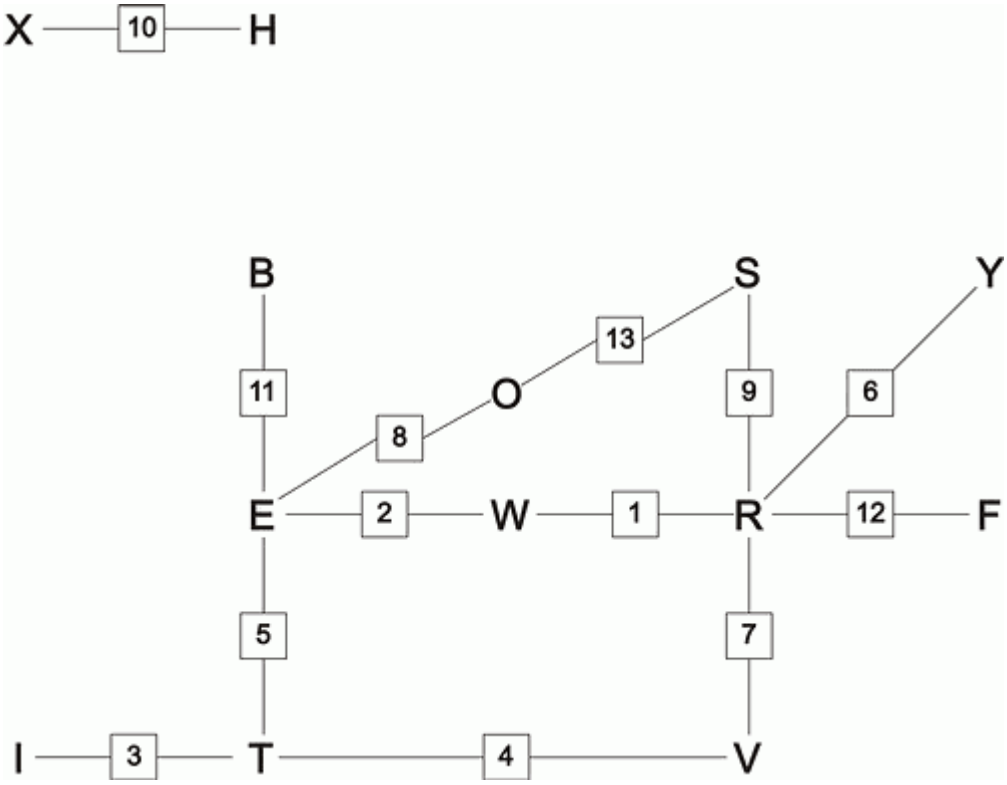


Figure 5.1 WETTERVORHERSAGEBISKAYA Split Menu 1

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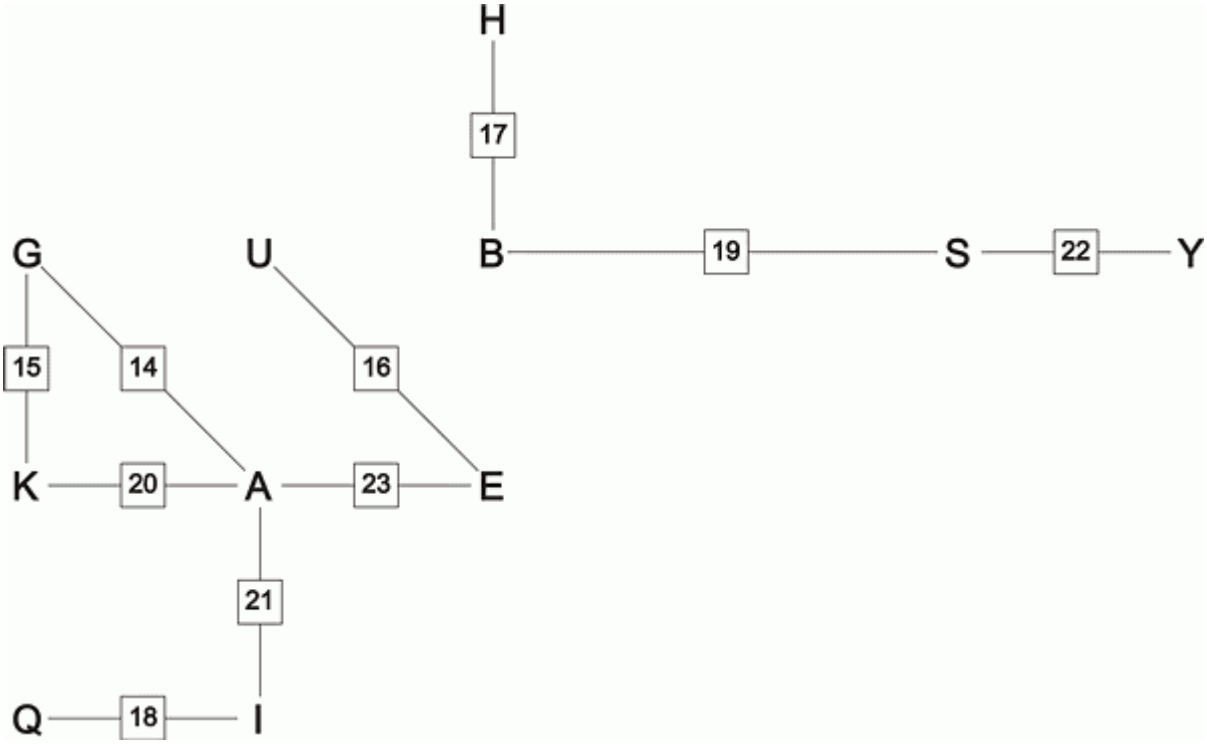


Figure 5.2 WETTERVORHERSAGEBISKAYA Split Menu 2

These new menus are represented in Figures 5.1 and 5.2. Figure 5.1 shows a stronger menu than Figure 5.2, containing as it does two loops in the main web of 11 letters and one additional web of 2 letters. Figure 5.2 shows a menu containing one loop in the main web of seven letters and one additional web of four letters. Neither of these menus is strong enough to produce a small number of stops which demonstrates the considerable difficulty of finding good menus even with an excellent crib like WETTERVORHERSAGEBISKAYA.

There are many aspects of the cryptanalytical operation that we have not considered in this brief essay:

We have not considered the great difficulty in predicting the frequencies on which messages would be transmitted, nor have we considered the difficulty in accurately intercepting and recording messages which may have been transmitted at a great distance from the interception stations often yielding a weak signal in poor atmospheric conditions.

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We have not considered the difficulty in composing cribs, or the frustration of having composed a crib and having matched it to the ciphertext only to discover it is of little use because it yields a feeble menu.

We have not discussed the valuable bombe time wasted running menus derived from false cribs, or a correct crib in the wrong position.

We have not considered the potential unreliability of the electromechanical components in the bombes, nor of the fact that the bombes were seldom available in numbers adequate to cope with the number of keys which burgeoned as the war progressed.

We have not considered the urgent necessity of breaking the messages while the intelligence they contained was of operational value.

We have not considered the difficulty of disseminating the intelligence to the field commanders without it being intercepted by the enemy, nor the difficulty of employing the intelligence operationally without arousing the enemy's suspicions that the Enigma was being broken.

We have not considered improvements to the Enigma that were being introduced as the war entered its final phase. The introduction of a pluggable *Umkehrwalze*, known as *Umkehrwalze D,* threatened the entire Enigma-breaking operation, but was never put into large scale use. The *Uhr*, a mechanical attachment to the Enigma, provided the Enigma with another layer of complexity. Doubly-enciphering a message rendered all cribs useless.

The Germans failed to avail themselves of a very simple technique which would have made the Enigma much more secure. They could have placed the occasional random letter in the plaintext before encipherment. This simple expedient would have made it very difficult to match cribs to the ciphertext and so would have greatly increased the difficulty of obtaining good menus. During the height of the Normandy campaign, Bletchley Park was deciphering a staggering 18,000 messages each day.

When Germany surrendered, Churchill ordered the destruction of every one of the 200 bombes which had broken the Enigma on this industrial scale. Not a single fragment remains.

In intellectual challenge and commitment of resources, the production of ULTRA is equalled only by the MANHATTAN project and putting a man on the Moon. Its effect on history ranks with either.